Review Chain Rule Partial Diff.

Given $w = r^2 + sv + t^3$

$r = x^2 + y^2 + z$

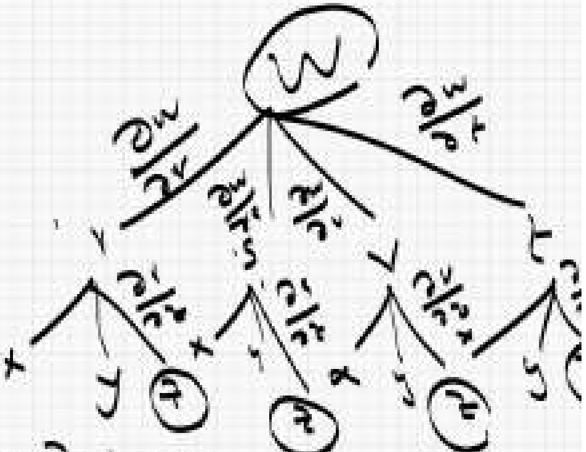
$s = xyz$

$v = x \cdot e^y$

$t = yz^2$

Find $\dfrac{\partial w}{\partial z} = w_z$

Sol.

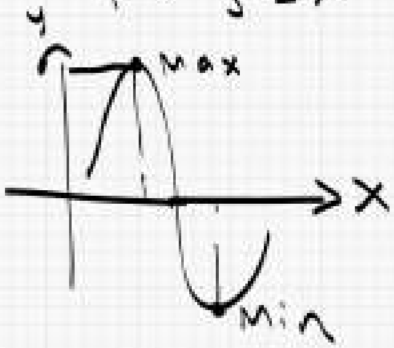


$$\frac{\partial w}{\partial z} = \frac{\partial w}{\partial r} \cdot \frac{\partial r}{\partial z} + \frac{\partial w}{\partial s} \cdot \frac{\partial s}{\partial z} + \frac{\partial w}{\partial v} \cdot \frac{\partial v}{\partial z} + \frac{\partial w}{\partial t} \cdot \frac{\partial t}{\partial z}$$

$$= (2r)(1) + (v)(xy) + (s)(0) + (3t^2)(2yz)$$

→ Lagrange multiplier (λ)

$y = f(x)$ Ex $y = x^2$



$y' = 0$

$x = \checkmark\checkmark$

⇒ multivariable

$z = F(x, y)$

$w = f(x, y, z)$

given $f(x, y, z)$

b/c Constraint $g(x, y, z) = k$

$\boxed{\text{Case 1}}$ $f(x, y)$ original

Given Constraint $g(x, y) = k$

$\boxed{\text{Step 1}}$ $F(x, y)$, $g(x, y)$

$\boxed{\text{Step 2}}$ Lagrange: $x, y, \lambda = ??$

$F_x = \lambda g_x \rightarrow ①$

$F_y = \lambda g_y \rightarrow ②$

Step 3

$g(x, y) = k \rightarrow ③$

Step 4 Solve 1, 2, 3 Find x, y, λ

Ex1 given $f(x,y) = 4xy$

Subject to $x^2+y^2=8$

find max, min value

SOL

Step 1 $f(x,y) = 4xy$

$g(x,y) = x^2+y^2$

Step 2 $\therefore f_x = \lambda g_x$

$\boxed{4y = \lambda \cdot 2x} \to (1)$

$\therefore f_y = \lambda g_y$

$\boxed{4x = \lambda \cdot 2y} \longrightarrow (2)$

Step 3 Solve:

$x^2+y^2 = 8 \longrightarrow (3)$

from (1), (2) $\lambda = \frac{4y}{2x} = \frac{4x}{2y}$

Step 4 $x^2 = y^2$

Sub in (3)

$x^2 + x^2 = 8 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$

$y = \pm 2$

Step 5

$F(2,2) = 4(2)(2) = 16$ max

$F(2,-2) = -16$ Min

$F(-2,2) = -16$

$F(-2,-2) = 16$

→ find max, min → lagrange

$$f(x,y) = x e^{y}$$

Subject to $x^2 + y^2 = 2$

SoL

Step ① $f(x,y) = x e^{y}$

$g(x,y) = x^2 + y^2$

Step ② $\because f_x = \lambda g_x$

$$e^{y} = \lambda \cdot 2x \rightarrow ①$$

$\because f_y = \lambda g_y$

$$x e^{y} = \lambda \cdot 2y \rightarrow ②$$

Step ③ Solve

$$x^2 + y^2 = 2 \rightarrow ③$$

$$\lambda = \frac{\cancel{e^{y}}}{2x} = \frac{x \cancel{e^{y}}}{2y} \Rightarrow \boxed{y = x^2}$$

Step 4

$$x^2 + x^4 = 2$$

$$x^4 + x^2 - 2 = 0$$

$$(x^2 - 1)(x^2 + 2) = 0$$

$x^2 = 1$ | $x^2 = -2$

$\boxed{x = \pm 1}$ | Reject

$y = 1$

Step 5

$F(1, 1) = e^{y}$ max

$F(-1, 1) = -e^{y}$ min

Case 2 given $\boxed{f(x,y,z)}$

subject to

one Constraint

$$g(x,y,z) = k$$

Step ① $f(x,y,z)$

$g_1(x,y,z)$

Step ②

$$F_x = \lambda g_x \to 1$$
$$F_y = \lambda g_y \to 2$$
$$F_z = \lambda g_z \to 3$$

Step ③ Solve ①, ②, ③

$$g(x,y,z) = k \to 4$$

Step ④ $x, y, z = ??$

Step ⑤ Subs. Max / Min

two Constraints

④ $\leftarrow g_1(x,y,z) = k_1$

⑤ $\leftarrow g_2(x,y,z) = k_2$

Step ① f, g_1, g_2

Step ②

$$F_x = \lambda g_{1x} + \mu g_{2x} \to ①$$
$$F_y = \lambda g_{1y} + \mu g_{2y} \to ②$$
$$F_z = \lambda g_{1z} + \mu g_{2z} \to ③$$

Step ③ solve ①, ②, ③, ④, ⑤

Step ④ $x, y, z = ??$

Subs.

Exe max $f(x,y,z) = 3x + y + 2z$

Subject to

$$y^2 + z^2 = 1$$
$$x + y - z = 1$$

Sol

Step ①

$$f(x,y,z) = 3x + y + 2z$$
$$g_1(x,y,z) = y^2 + z^2$$
$$g_2(x,y,z) = x + y - z$$

Step 2

$$F_x = \lambda g_{1x} + \mu g_{2x}$$
$$3 = \cancel{\lambda(0)} + \mu(1) \Rightarrow \boxed{\mu = 3} \to ①$$
$$F_y = \lambda g_{1y} + \mu g_{2y}$$
$$1 = \lambda \cdot 2y + \cancel{\mu} \cdot 1 \Rightarrow 2y = -1 \to ②$$

$f_{z} = \lambda g_{1z} + \mu g_{2z}$

$$\boxed{\lambda z = \frac{5}{2}} \rightarrow (3)$$

Solve!

$\mu = 3 \rightarrow (1)$

$\lambda y = -1 \rightarrow (2)$

$\lambda z = \frac{5}{2} \rightarrow (3)$

$y^2 + z^2 = 1 \rightarrow (4)$

$x + y - z = 1 \rightarrow (5)$

$y = \frac{-1}{\lambda}$

$z = \frac{5}{2\lambda}$

$$\boxed{\lambda = \pm \frac{\sqrt{29}}{2}}$$

$\lambda = \frac{\sqrt{29}}{2}$

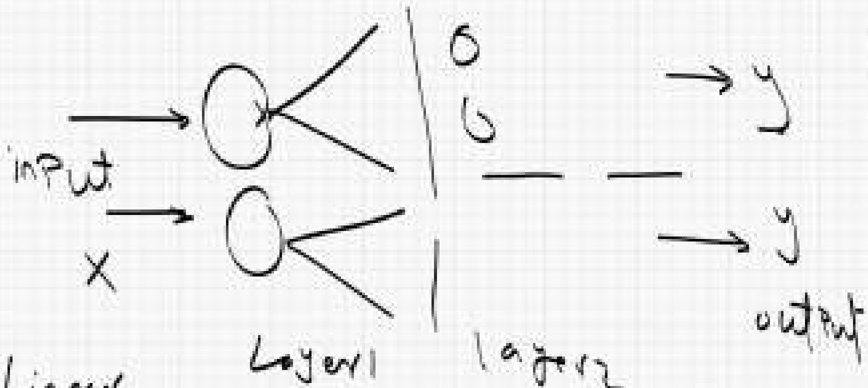
$y, z, x = \checkmark$

$\lambda = -\frac{\sqrt{29}}{2}$

x, y, z

Chain Rule

Backpropagation



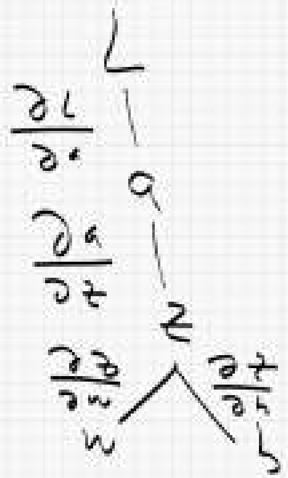
linear
$\longrightarrow \quad z = wx + b \rightarrow$ ①

$\longrightarrow \quad a = \sigma(z) = \dfrac{1}{1+e^{-z}} \rightarrow$ ②
activation

$\rightarrow L = \frac{1}{2}(a-y)^2 \longrightarrow$ ③ Loss

Chain Rule $\dfrac{\partial L}{\partial w}, \dfrac{\partial L}{\partial b} = ??$

Note $x, y \rightarrow$ given



? ? ?

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial b} \leftarrow ?$$

$$\longrightarrow z = wx + b$$

$$a = \frac{1}{1+e^{-z}}$$

$$L = \frac{1}{2}(a-y)^2$$

$x, y \to$ given

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$$= \frac{1}{\cancel{2}} \cdot \cancel{2}(a-y) \cdot \boxed{\frac{\partial a}{\partial z}} \cdot x$$

$$\therefore \frac{\partial a}{\partial z} = \frac{\cancel{(1+e^{-z})\,0} - 1(-e^{-z})}{(1+e^{-z})^2}$$

$$\frac{\partial a}{\partial z} = \frac{e^{-z}}{(1+e^{-z})^2} = \boxed{a(1-a)} \ \#$$

$$\frac{1}{1+e^{-z}}\left(1 - \frac{1}{1+e^{-z}}\right)$$

$$\frac{1}{1+e^{-z}}\left(\frac{\cancel{1}+e^{-z}\cancel{-1}}{1+e^{-z}}\right)$$

$$= \frac{e^{-z}}{(1+e^{-z})^2}$$

$$\boxed{\frac{\partial L}{\partial w} = (a-y)a(1-a)x} \ \#$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial b}$$

$$\boxed{= (a-y) \cdot a(1-a) \cdot 1} \ \#$$

Steps of Solution
Backpropagation

Step ① given → $z = wx + b$

or $z = w_1 x_1 + w_2 x_2 + b$

→ $a = \frac{1}{1+e^{-z}}$

→ $L = \frac{1}{2}(a-y)^2$

Step ② use given values

$z = \checkmark$
$a = \checkmark$
$L = \checkmark$

Step ③ Chain Rule

$$\left.\begin{aligned} \frac{\partial L}{\partial w} &= \checkmark \\ \frac{\partial L}{\partial b} &= \checkmark \end{aligned}\right\} \text{values}$$

Step ④ update Parameters

$$w^{new} = w - \alpha \frac{\partial L}{\partial w}$$

$$b^{new} = w - \alpha \frac{\partial L}{\partial b}$$

learning rate

Exe lecture

Given single neuron

$$z = wx + b$$

$$a = \sigma(z) = \frac{1}{1+e^{-z}}$$

$$L = \frac{1}{2}(a-y)^2$$

Use the values

$x = 1$, $y = 1$, initial $w = 0$, $b = 0$)

Learning rate $\alpha = 0.1$

Perform full training step

Sol

Step ①

$$z = 0 \cdot 1 + 0 = 0$$

$$a = \frac{1}{1+e^{0}} = \frac{1}{2} = 0.5$$

$$L = \frac{1}{2}(0.5-1)^2 = \boxed{0.125}$$

Step ② Chain Rule.

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$$= (a-y)a(1-a) \cdot x$$

$$= (0.5-1)0.5(1-0.5) \cdot 1 = -0.125$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial b}$$

$$= (a-y)a(1-a) \cdot 1 = -0.125$$

$$w^{new} = w - \alpha \frac{\partial L}{\partial w}$$

$$= 0 - 0.1(-0.125)$$

$$= \boxed{0.0125}$$

$$b^{new} = b - \alpha \frac{\partial L}{\partial b}$$

$$= 0 - 0.1(-0.125)$$

$$= \boxed{0.0125}$$

lect

Problem3 Single neuron

$$z = w_1 x_1 + w_2 x_2 + b$$

$$a = \frac{1}{1+e^{-z}}$$

$$L = \frac{1}{2}(a-y)^2$$

Given data $[x_1 = 1, x_2 = 2, y = 1]$

$w_1 = 0.1$, $w_2 = 0.2$, $b = 0.3$

$\alpha = 0.1$

Sol.

Step I $\quad z = w_1 x_1 + w_2 x_2 + b$

$$z = 0.8$$
$$a = 0.68997$$
$$L = 0.04806$$

Step 2 Chain

$$\frac{\partial L}{\partial w_1} = \underbrace{\frac{\partial L}{\partial a}}_{(a-y)} \cdot \underbrace{\frac{\partial a}{\partial z}}_{a(1-a)} \cdot \underbrace{\frac{\partial z}{\partial w_1}}_{x_1}$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \underbrace{\frac{\partial z}{\partial w_2}}_{x_2}$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial a} \frac{\partial a}{\partial z} \cdot \underbrace{\frac{\partial z}{\partial b}}_{1}$$

$$\frac{\partial L}{\partial w_1} = -0.06632$$

$$\frac{\partial L}{\partial w_2} = -0.1326$$

$$\frac{\partial L}{\partial b} = -0.06632$$

Step update Para.

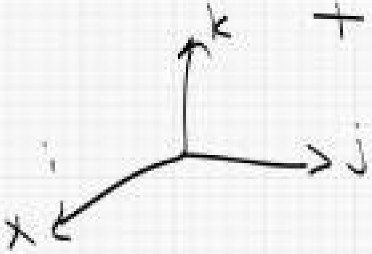
$$w_1^{new} = w_1 - \alpha \frac{\partial L}{\partial w_1} \quad \checkmark$$

$$w_2^{new} = w_2 - \alpha \frac{\partial L}{\partial w_2} \quad \checkmark$$

$$b^{new} = b - \alpha \frac{\partial L}{\partial b} \quad \checkmark$$

$\boxed{1st}$ Gradient

$$\nabla = \frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k$$



$$\nabla F = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j + \frac{\partial f}{\partial z} k$$

$$= \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

Max rate of change
or min " " "

$\longrightarrow$ Max rate of change at Point (a,b)

$$= \| \nabla F(a,b) \|$$

$$= \sqrt{(\ )^2 + (\ )^2 + \cdots}$$

Direction $u = \dfrac{\nabla F(a,b)}{\| \nabla F(a,b) \|}$

Min $\longrightarrow$ Add $(\underline{-})$ Sign

$\longrightarrow f(x,y) = x^2 + y^2$

at Point $(1,3)$

Sol

Max rate of chan

$= \| \nabla F(1,3) \|$

$\nabla f = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j$

$= \langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \rangle$

$= \langle 2x, 2y \rangle$

$\nabla f(\overset{x}{1}, \overset{y}{3}) = \langle 2, 6 \rangle$

Max rate $= \| \nabla f(1,3) \|$

$= \sqrt{(2)^2 + (6)^2} = \sqrt{40}$

Direction $u = \frac{\nabla f(1,3)}{\| \nabla f(1,3) \|}$

$= \frac{\langle 2, 6 \rangle}{\sqrt{40}}$

$= \frac{2i + 6j}{\sqrt{40}}$

$= \frac{2}{\sqrt{40}} i + \frac{6}{\sqrt{40}} j$

l'Hopital

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{0}{0} \Big| \frac{\infty}{\infty}$$

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

$\frac{0}{0} \Big| \frac{\infty}{\infty}$ = value

$$= \lim_{x \to a} \frac{f''(x)}{g(x)}$$

Ex $\lim_{x \to 0} \frac{\sin x}{x} = \frac{0}{0}$

$$\lim_{x \to 0} \frac{\cos x}{1} = \cos 0 = 1$$

Ex $\lim_{x \to 0^+} \sqrt{x} \cdot \ln x = 0 \cdot (-\infty) \quad \left(\frac{0}{0} \Big| \frac{\infty}{\infty} \right)$

$$\lim_{x \to 0} \frac{\ln x}{\frac{1}{\sqrt{x}}} \; x^{\frac{1}{2}} = \frac{\infty}{\infty}$$

$$= \frac{1}{0} = \infty$$

L

$$\lim_{x \to 0} \frac{\frac{1}{x}}{\frac{1}{2} x^{-\frac{3}{2}}}$$

Case

$\boxed{\infty - \infty}$

$\rightarrow \left(\frac{0}{0} \middle| \frac{\infty}{\infty} \right)$

Ex $\lim_{x \to 0} \left(\frac{1}{x} - \csc x \right)$

Sol

$= \infty - \infty$

$= \lim_{x \to 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$

$= \lim_{x \to 0} \left(\frac{\sin x - x}{x \sin x} \right)$

$= \boxed{\frac{0}{0}}$

$\lim_{x \to 0} \frac{\cos x - 1}{x \cdot \cos x + \sin x \cdot 1}$